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In [1]: # Behzad Ghabaei
# CS 82A Data Science
# module6_lab3.ipynb
# Textbook Appendix C: Retrieve and Search
# C.3 - C.4
# Instructor Seno
# 10/7/2026
```

```
In [2]: import sqlite3
import pandas as pd

con = sqlite3.connect("database.db")
```

```
In [3]: create_table_sql = """
CREATE TABLE IF NOT EXISTS customers (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    name TEXT NOT NULL,
    city TEXT,
    credit_limit REAL
);
"""
```

```
In [4]: insert_rows_sql = """
INSERT OR REPLACE INTO customers (id, name, city, credit_limit) VALUES
(1, 'Ali Hussein', 'New York', 5000.00),
(2, 'Bobby Jones', 'Los Angeles', 3500.50),
(3, 'John Brown', 'Chicago', 2500.00),
(4, 'Billy Prince', 'Boston', 7500.00),
(5, 'Eve Kim', 'Austin', 4200.00),
(6, 'Linda James', 'Chicago', 1500.75),
(7, 'Cathy Clark', 'Miami', 6000.00),
(8, 'Hilda Bosche', 'Seattle', 3100.00),
(9, 'Cynthia Bush', 'Miami', 3200.00),
(10, 'Cathy Jones', 'Miami', 3400.00),
(11, 'Annie Smith', 'Miami', 2200.00),
(12, 'Casey Blanc', 'Miami', 1200.00),
(13, 'Catalina Caballo', 'Miami', 1200.00);
"""
```

Here the "text_search_using_LIKE" variable returns all the records in the "customers" table that were from the "city" of "Miami" and have letters like "Ca" in their "name" which are ordered by the "credit_limit" they carry on their credit card. The table should return only Miami residents that have Ca in their name such as Catalina, Casey, and Cathy, but not Cynthia or Annie. SQLBolt.com can help to learn the use of the LIKE keyword. (sqlbolt.com | "SQL Lesson 3: Queries with constraints (Pt. 2)" | https://sqlbolt.com/lesson/select_queries_with_constraints_pt_2)

```
In [5]: text_search_using_LIKE = """
SELECT *
FROM customers
WHERE city == "Miami"
AND name LIKE "%Ca%"
ORDER BY credit_limit;

"""
```

```
In [6]: df_chart = pd.read_sql_query("SELECT * FROM customers;", con)
df_chart
```

Out[6]:

	id	name	city	credit_limit
0	1	Ali Hussein	New York	5000.00
1	2	Bobby Jones	Los Angeles	3500.50
2	3	John Brown	Chicago	2500.00
3	4	Billy Prince	Boston	7500.00
4	5	Eve Kim	Austin	4200.00
5	6	Linda James	Chicago	1500.75
6	7	Cathy Clark	Miami	6000.00
7	8	Hilda Bosche	Seattle	3100.00
8	9	Cynthia Bush	Miami	3200.00
9	10	Cathy Jones	Miami	3400.00
10	11	Annie Smith	Miami	2200.00
11	12	Casey Blanc	Miami	1200.00
12	13	Catalina Caballo	Miami	1200.00

LIKE keyword returns these 4 records, with "Ca" in their name, (using "% %" quotes and the percent sign.) WHERE keyword returns these records who are people who reside in Miami. The credit limit is sorted, least to greatest with the key words ORDER BY. WHERE uses the AND keyword with LIKE. The list returns: Casey Blanc, Catalina Caballo, Cathy Jones, and Cathy Clark (Cathy Clark has the highest credit limit)

```
In [7]: df_query = pd.read_sql_query( text_search_using_LIKE, con )
df_query
```

Out[7]:

	id	name	city	credit_limit
0	12	Casey Blanc	Miami	1200.0
1	13	Catalina Caballo	Miami	1200.0
2	10	Cathy Jones	Miami	3400.0
3	7	Cathy Clark	Miami	6000.0

% is a wildcard character used for any string of zero or more characters, for example WHERE city LIKE '%Los Angeles%' finds all cities with Los Angeles anywhere in the city column.

```
In [8]: sql_ex = """
SELECT * FROM customers
WHERE city
LIKE "%Los Angeles%";
"""

done_ex = pd.read_sql_query(sql_ex, con)
done_ex
```

Out[8]:

	id	name	city	credit_limit
0	2	Bobby Jones	Los Angeles	3500.5

```
In [9]: con.execute(create_table_sql)
con.execute(insert_rows_sql)
```

```
con.execute(text_search_using_LIKE)
```

Out[9]: <sqlite3.Cursor at 0x2d76956d140>

```
In [10]: con.commit()
```

```
In [11]: con.close()
```

Behzad Ghabaei - End Of Lab

In []: